

Q1 Find your parameter from your uni in the Fall2025ClassList.xlsx file

```
Out[4]: Name           Mailloux, Brian
WellData           Platform
Site              SiteF
AsSpecies          As2S3
NYCTree            guards
Core#              18NYHC044
Poster Group       3
Name: bjm2103, dtype: object
```

Q2 Read NYC Tree data

- print columns
- print .info()

check columns

```
Out[9]: Index(['tree_id', 'block_id', 'created_at', 'tree_dbh', 'stump_diam',
              'curb_loc', 'status', 'health', 'spc_latin', 'spc_common', 'steward',
              'guards', 'sidewalk', 'user_type', 'problems', 'root_stone',
              'root_grate', 'root_other', 'trunk_wire', 'trnk_light', 'trnk_other',
              'brch_light', 'brch_shoe', 'brch_other', 'address', 'postcode',
              'zip_city', 'community board', 'borocode', 'borough', 'cnclldist',
              'st_assem', 'st_senate', 'nta', 'nta_name', 'boro_ct', 'state',
              'latitude', 'longitude', 'x_sp', 'y_sp', 'council district',
              'census tract', 'bin', 'bbl'],
              dtype='object')
```

Do info and make sure things look ok

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 683788 entries, 0 to 683787
Data columns (total 45 columns):
#   Column                Non-Null Count  Dtype
---  -
0   tree_id                683788 non-null  int64
1   block_id               683788 non-null  int64
2   created_at             683788 non-null  object
3   tree_dbh               683788 non-null  int64
4   stump_diam             683788 non-null  int64
5   curb_loc               683788 non-null  object
6   status                 683788 non-null  object
7   health                 652172 non-null  object
8   spc_latin              652169 non-null  object
9   spc_common             652169 non-null  object
10  steward                652173 non-null  object
11  guards                 652172 non-null  object
12  sidewalk               652172 non-null  object
13  user_type              683788 non-null  object
14  problems                652124 non-null  object
15  root_stone             683788 non-null  object
16  root_grate             683788 non-null  object
17  root_other             683788 non-null  object
18  trunk_wire             683788 non-null  object
19  trnk_light             683788 non-null  object
20  trnk_other             683788 non-null  object
21  brch_light             683788 non-null  object
22  brch_shoe              683788 non-null  object
23  brch_other             683788 non-null  object
24  address                683788 non-null  object
25  postcode               683788 non-null  int64
26  zip_city               683788 non-null  object
27  community board        683788 non-null  int64
28  borocode               683788 non-null  int64
29  borough                683788 non-null  object
30  cncldist               683788 non-null  int64
31  st_assem               683788 non-null  int64
32  st_senate              683788 non-null  int64
33  nta                    683788 non-null  object
34  nta_name               683788 non-null  object
35  boro_ct                683788 non-null  int64
36  state                  683788 non-null  object
37  latitude                683788 non-null  float64
38  longitude               683788 non-null  float64
39  x_sp                   683788 non-null  float64
40  y_sp                   683788 non-null  float64
41  council district       677269 non-null  float64
42  census tract           677269 non-null  float64
43  bin                    674229 non-null  float64
44  bbl                    674229 non-null  float64
dtypes: float64(8), int64(11), object(26)
memory usage: 234.8+ MB

```

Q3 What is the description of of your parameter from the data dictionary

From the data dictionary Guards means the Presence and type of tree guard. Values Harmful, Helpful, and Unsure all indicate that a tree guard is present. A tree guard is considered 'helpful' if it doesn't impede water getting to the tree and does not raise the soil level or trap debris in the pit.

Q4 report a count of the breakdown of your parameter

```
Out[8]: guards
None      572306
Helpful    51866
NaN        31616
Harmful    20252
Unsure     7748
Name: count, dtype: int64
```

Q5 convert the count to a percentage

```
Out[11]: guards
None      83.696409
Helpful    7.585099
NaN        4.623655
Harmful    2.961737
Unsure     1.133100
Name: proportion, dtype: float64
```

Q6 How does DBH vary with your parameter?

- as a table

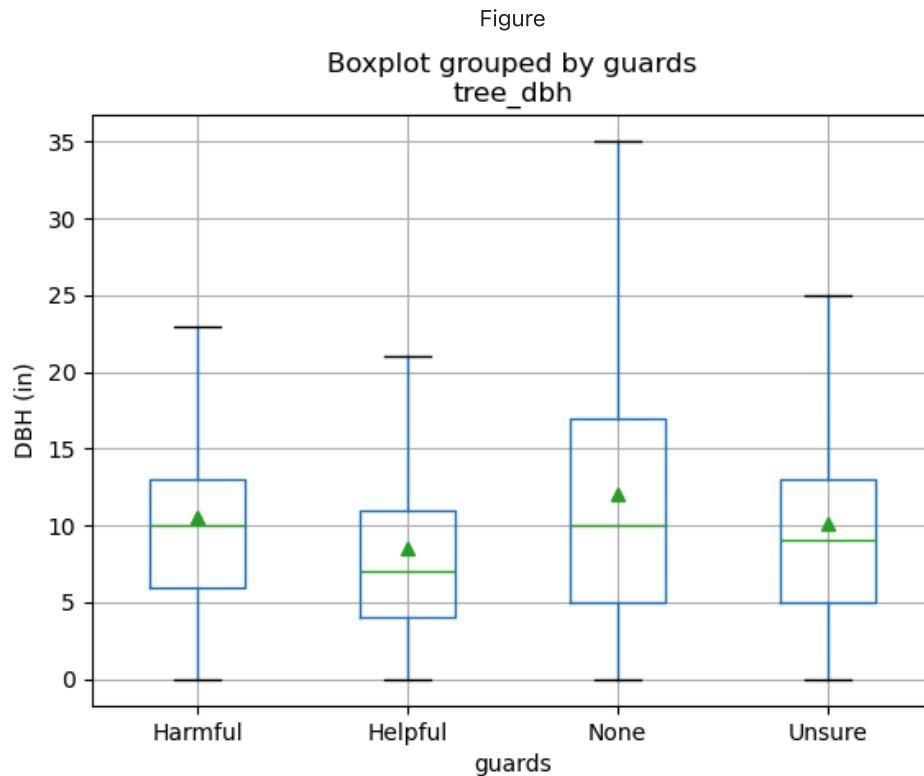
```
Out[13]:
```

	count	mean	std	min	25%	50%	75%	max
guards								
Harmful	20252.0	10.583646	6.080190	0.0	6.0	10.0	13.0	170.0
Helpful	51866.0	8.537886	6.267371	0.0	4.0	7.0	11.0	159.0
None	572306.0	12.058201	8.851880	0.0	5.0	10.0	17.0	425.0
Unsure	7748.0	10.125452	6.757314	0.0	5.0	9.0	13.0	53.0
NaN	31616.0	2.415992	5.050724	0.0	0.0	0.0	3.0	450.0

As a boxplot

- show the mean
- show no outliers

```
Out[27]: Text(0, 0.5, 'DBH (in)')
```



Q7 Summarize

Interestingly, bigger trees have harmful or no guards.

At fist this seems counterintuitive but perhaps as trees grow larger they start to encroach on the guards. Thus correlation is not causation.

Q8 Compare the total counts of your parameter across boroughs

- As a table

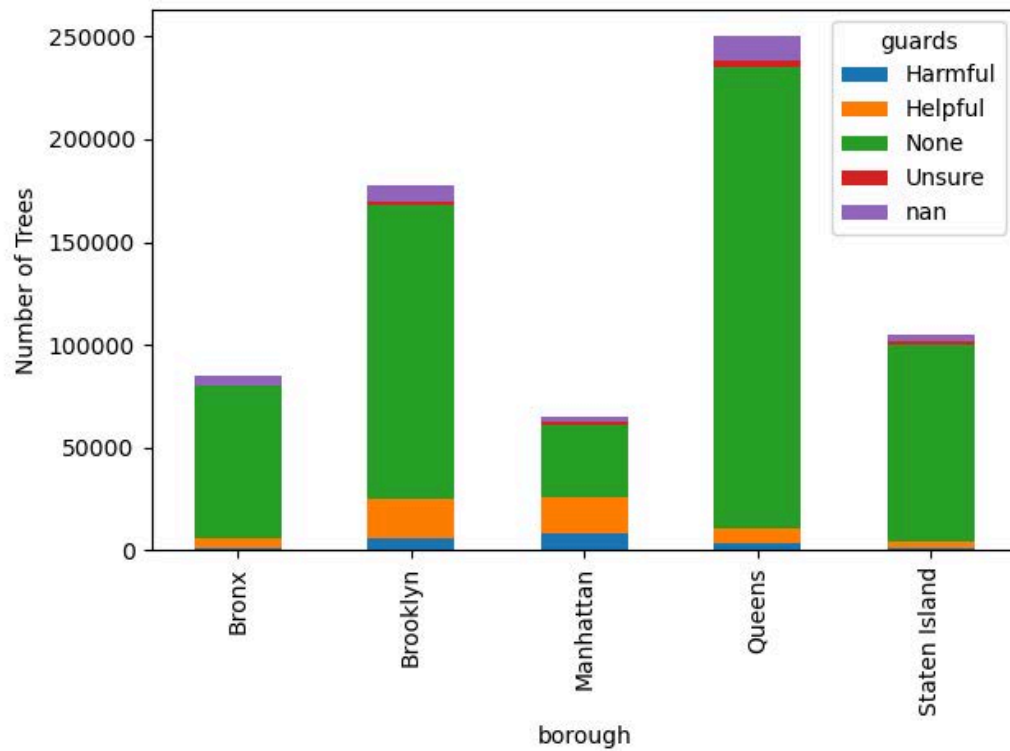
Out[19]:

guards	Harmful	Helpful	None	Unsure	NaN
borough					
Bronx	976	4605	74524	480	4618
Brooklyn	5650	19471	142563	2060	7549
Manhattan	8741	17559	34812	1315	2996
Queens	3690	7153	224356	2774	12578
Staten Island	1195	3078	96051	1119	3875

- as a bar chart

Out[28]: Text(0, 0.5, 'Number of Trees')

Figure



Q9 convert to a percentage by borough

- as a table

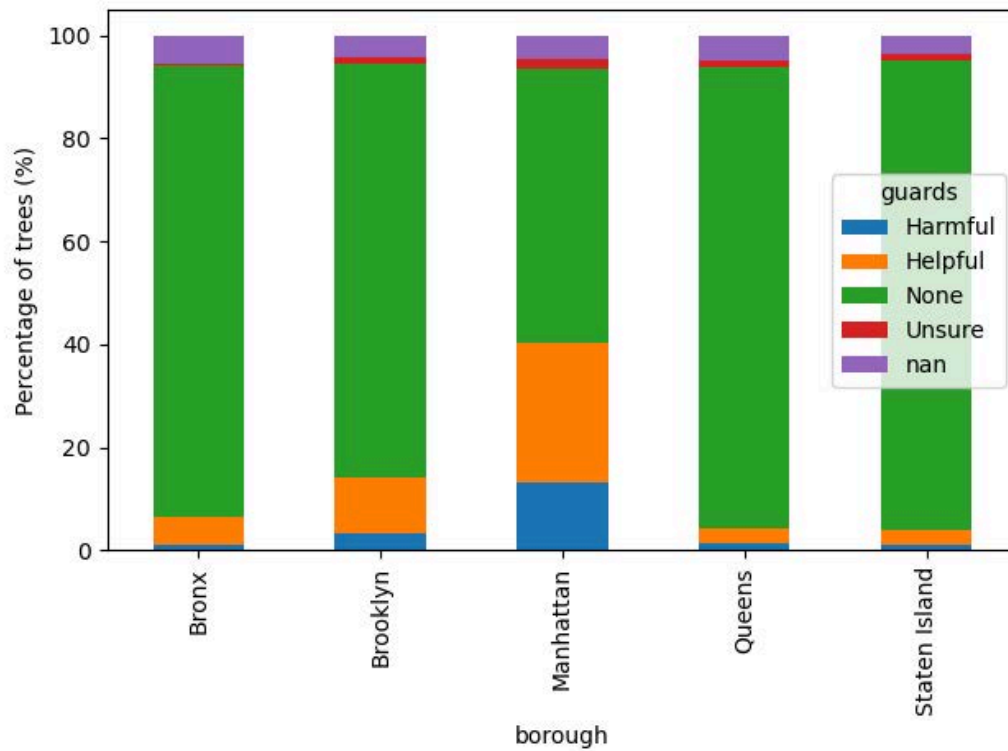
Out[22]:

guards	Harmful	Helpful	None	Unsure	NaN
borough					
Bronx	1.145500	5.404739	87.466404	0.563360	5.419997
Brooklyn	3.186815	10.982385	80.410958	1.161918	4.257923
Manhattan	13.360745	26.839185	53.210645	2.009996	4.579429
Queens	1.472754	2.854908	89.545043	1.107160	5.020136
Staten Island	1.134659	2.922577	91.200934	1.062496	3.679333

as a bar chart

Out[29]: Text(0, 0.5, 'Percentage of trees (%)')

Figure



Q10 summarize

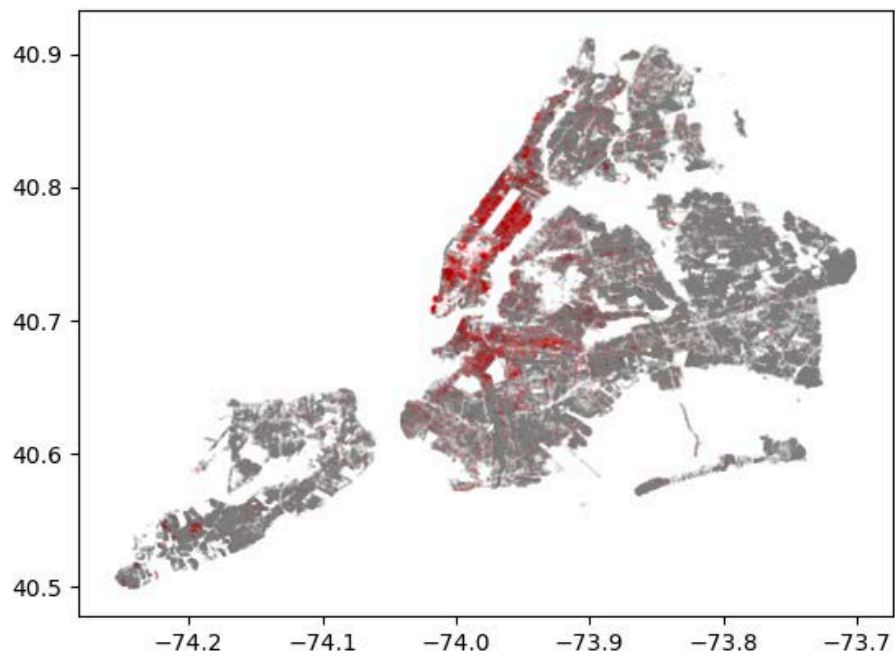
Brooklyn and Queens has the most trees whereas Brooklyn and Manhattan have the most trees with guards. But Manhattan has by far the highest percentage of trees with guards, both helpful and harmful.

Q11 make a quick simple map to really bring it home!

- no legend needed.
- choose a simple color scheme for your parameter

```
Out[24]: <matplotlib.collections.PathCollection at 0x1421a4610>
```

Figure



First Bonus 5%

- Post a selfie with a tree that relates to your parameter or is interesting.
- I didn't have a selfie when making the homework. I like the cobblestones and guard.

Out[32]:



Q Final 5%

Do a t-test or anova with your parameter

F value: 4178.336, p value: 0.000

Multiple Comparison of Means – Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
Bronx	Brooklyn	2.0452	0.0	1.9472	2.1432	True
Bronx	Manhattan	-1.22	0.0	-1.3422	-1.0978	True
Bronx	Queens	2.8642	0.0	2.771	2.9575	True
Bronx	Staten Island	0.7991	0.0	0.6908	0.9074	True
Brooklyn	Manhattan	-3.2652	0.0	-3.3728	-3.1577	True
Brooklyn	Queens	0.819	0.0	0.746	0.8919	True
Brooklyn	Staten Island	-1.2461	0.0	-1.3376	-1.1547	True
Manhattan	Queens	4.0842	0.0	3.981	4.1874	True
Manhattan	Staten Island	2.0191	0.0	1.9021	2.1361	True
Queens	Staten Island	-2.0651	0.0	-2.1515	-1.9788	True